Mini express service

modifies history

Version	Contributor	Date	Change Log
v1.0	logic.peng	18/08/31	init

Summary

Mini service is a service application in ROBOT. At present, Mini's eyes are two LCD screens. By rendering GIF or Lottie animation on the screen, Mini can display predefined expressions given by the design. This document briefly describes these interfaces and SDK integration

SDK

- protobuf struct

```
enum ExpressFormat {
    GIF = 0;
    PNG = 1;
}

//epxress info
message ExpressInfo {
    int32 id = 1; //id
    string name = 2; //name
```

```
    int32 duration = 3; //duration
    int32 frames = 5; //frames
    ExpressFormat format = 4;//json, gif, png
    int32 intrinsicWidth = 6; //
    bool customize = 7;//
}
```

- java interface

```
public class ExpressApi {
 /**
  * Get the support expression on the robot
  *
  * @return list of Express.ExpressInfo
  */
 public @Nullable List<Express.ExpressInfo> getExpressList() ;

 /**
  * Getting facial expression animation on /sdcard/customize/sdcard on the robot
  *
  * @return list of {@link Express.ExpressInfo}
  */
 public List<Express.ExpressInfo> getCustomizeExpressList() ;

 /**
  * do a express
  *
  * @param name express file name
  */
 public void doExpress(@NonNull String name) ;

 /**
  * do a express
  *
```

```
     * @param name express file name
     * @param loopCount loop
     */
    public void doExpress(String name, @IntRange(from = 0, to = Character.MAX_VALUE) int loopCount) ;

    /**
     * do a express
     *
     * @param name express file name
     * @param priority priority
     * @param loopCount loop
     */
    public void doExpress(String name, @IntRange(from = 0, to = Character.MAX_VALUE) int loopCount,
        Priority priority) ;

    /**
     * do a express
     *
     * @param name express file name
     * @param loopCount loop
     * @param tweenable tweenable
     * @param priority priority {@link Priority}
     */
    public void doExpress(String name, @IntRange(from = 0, to = Character.MAX_VALUE) int loopCount,
        boolean tweenable, Priority priority) ;

    /**
     * Do expression animation, and monitor the expression animation state
     *
     * @param name express file name
     * @param loopCount loop
     * @param priority priority {@link Priority}
     * @param listener listener
     */
    public void doExpress(String name,
        @IntRange(from = 0, to = Character.MAX_VALUE) final int loopCount, Priority priority,
        @Nullable final AnimationListener listener) ;
```

```
/**
 * Do expression animation, and monitor the expression animation state
 *
 * @param name express file name
 * @param loopCount loop
 * @param tweenable tweenable
 * @param priority priority {@link Priority}
 * @param listener listener
 */
public void doExpress(final String name,
    @IntRange(from = 0, to = Character.MAX_VALUE) final int loopCount, final boolean tweenable,
    final Priority priority, @Nullable final AnimationListener listener) ;

/**
 * Do a built-in facial expression animation built into robot sdcard/customize/expresss dir
 *
 * @param name express file name
 * @param priority priority
 * @param listener  {@link AnimationListener}
 */
public void doCustomizeExpress(final String name, Priority priority,
    @Nullable final AnimationListener listener) ;

/**
 * Do a built-in facial expression animation built into robot sdcard/customize/expresss dir
 *
 * @param name express file name
 * @param priority priority
 * @param loopCount loop
 * @param listener  {@link AnimationListener}
 */
public void doCustomizeExpress(final String name, final int loopCount, Priority priority,
    @Nullable final AnimationListener listener) ;

/**
 * Do a built-in facial expression animation built into robot sdcard/customize/expresss dir
```

```
     *
     * @param name express file name
     * @param priority priority
     * @param tweenable tweenable
     * @param loopCount loop
     * @param listener  {@link AnimationListener}
     */
    public void doCustomizeExpress(final String name, final int loopCount, final boolean tweenable,
        Priority priority, @Nullable final AnimationListener listener) ;

    /**
     * Do expression animation, and monitor the expression animation state
     *
     * @param name express file name
     * @param listener listener
     */
    public void doExpress(String name, @Nullable final AnimationListener listener) ;

    /**
     * PC simulation call to display the specified frame of expression
     *
     * @param name file name
     * @param frame frame
     */
    public void setFrame(String name, @IntRange(from = 0) int frame) ;

    /**
     * PC simulation call to display the specified frame of expression
     *
     * @param name file name
     * @param frame frame
     * @param customize customize
     */
    public void setFrame(String name, @IntRange(from = 0) int frame, boolean customize) ;

    /**
     * Circular progress bar animation expression
```

```
     *
     * @param status 0 uncharging ,1 charging
     * @param progress progress
     * @param priority priority, {@link Priority}
     * @param animated animate enable
     * @param listener listener
     */
    public void doProgressExpress(@IntRange(from = 0, to = 1) final int status,
        final @IntRange(from = 0, to = 100) int progress, final boolean animated, Priority priority,
        @Nullable final AnimationListener listener);

    /**
     * Stop specifying the expression of file name. If the expression animation ends, nothing is done
     *
     * @param express express
     */
    public void stopExpress(@NonNull String express) ;
}


public interface AnimationListener {
    /**
     * Notifies the start of the animation.
     */
    void onAnimationStart();

    /**
     * Notifies the end of the animation.
     *
     * @param code error code
     */
    void onAnimationEnd(int code);

    /**
     * @param loopNumber loopNumber
     */
    void onAnimationRepeat(int loopNumber);
```

```
}
```